

An unbounded-distance genus-one obstruction to geodesic repair of permutations

Automated mathematics run `fa_banach_001`, lane 13

19 July 2026

Status. Counterexample likely valid, subject to human review. The construction disproves the full-cycle question as printed, not merely its multi-cycle extension.

1 Source question

The question appears in Cécilia Lancien's thesis [1], Section 9.16.3, page 148. With $\gamma = (1\ 2\ \cdots\ p)$ and Cayley distance $d(\sigma, \tau) = p - \#(\sigma^{-1}\tau)$, it asks whether there are universal constants θ, κ such that

$$d(\text{id}, \alpha) + d(\alpha, \gamma) = d(\text{id}, \gamma) + 2\delta$$

implies the existence of a geodesic permutation α' satisfying

$$d(\alpha, \alpha') = 2\delta', \quad \delta' \leq \theta\delta,$$

with a choice map $\alpha \mapsto \alpha'$ having fibres at most $p^{\kappa\delta}$. The exact source passage is reproduced below.

Remark 9.16.5. *The upper bound we established in Lemma 9.16.4 is probably far from optimal (e.g. it is likely that the exponent 4δ in the polynomial pre-factor in k and p can be improved). But this does not really matter for our specific goal. Nonetheless, in the special case of non-geodesic permutations between id and γ on $\{1, \dots, p\}$, it is in fact quite easy to obtain an upper bound which scales as $p^{3\delta}$ for the ratio between the number of 2δ non-geodesic permutations with a given number of cycles and the number of geodesic permutations with the same number of cycles. We present the result in Lemma 9.16.6 below, the problem being that the proof method does not seem to generalize so straightforwardly to the case that we truly need, that is the one of non-geodesic permutations between id and γ_f .*

Note also that very similar looking upper bounds had previously been derived regarding the cardinality of the set $\mathfrak{S}_\delta(p) = \{\alpha \in \mathfrak{S}(p) : \#(\gamma^{-1}\alpha) + \#(\alpha) = p + 1 - 2\delta\}$, which is the union of the sets $\mathfrak{S}_{\delta,m}(p)$ defined in Lemma 9.16.6 for $1 \leq m \leq p$. In particular, it was established in [142], Lemma 12, that for any $0 \leq \delta \leq \lfloor p/2 \rfloor$,

$$|\mathfrak{S}_\delta(p)| \leq |\mathfrak{S}_0(p)| p^{3\delta} = \text{Cat}_p p^{3\delta}.$$

However, this is definitely even less enough for our purpose: the latter really requires an upper bound on the number of permutations which have a given defect and a given number of cycles in terms of the number of permutations which have no defect and the same (or a related) number of cycles.

On the other hand, one may have hoped for a stronger result than these simply counting ones. For instance something like

$$d(\text{id}, \alpha) + d(\alpha, \gamma) = d(\text{id}, \gamma) + 2\delta \Rightarrow \exists \alpha' : d(\alpha, \alpha') = 2\delta' \text{ and } d(\text{id}, \alpha') + d(\alpha', \gamma) = d(\text{id}, \gamma),$$

with $\delta' \leq \theta\delta$ and with the mapping $\phi : \alpha \mapsto \alpha'$ satisfying $|\phi^{-1}(\alpha')| \leq p^{\kappa\delta}$, for some coefficients θ, κ . However, determining whether this kind of statement holds or not seems to remain an open question.

Lemma 9.16.6. *Let $p \in \mathbb{N}$ and denote by γ the canonical full cycle on $\{1, \dots, p\}$. For any $0 \leq \delta \leq \lfloor p/2 \rfloor$ and*

The interval $[\text{id}, \gamma]$ is the set of noncrossing permutations in the cyclic order $1, \dots, p$; see, for example, Hall [2, Theorem 3.1].

2 Idea of the proof

Keep the defect equal to one while making the obstruction larger. On $4m$ cyclically ordered points, take two families of m nested chords. Every chord in the first family crosses every chord in the second, so their intersection graph is $K_{m,m}$. The corresponding fixed-point-free involution has genus one for every m .

If a permutation is changed by r transpositions, the relative permutation is supported on at most $2r$ points. Consequently at least $2m - 2r$ of the original chord transpositions remain literal cycles. A noncrossing permutation cannot retain chords from both sides of the $K_{m,m}$ crossing pattern, hence it retains at most m . Therefore $r \geq m/2$, although the defect never exceeds one.

3 Counterexample family

For $m \geq 1$, put $p = 4m$, let $\gamma = (1\ 2 \cdots 4m)$, and define

$$\alpha_m = \prod_{i=1}^m (i, 3m+1-i) \prod_{j=1}^m (m+j, 4m+1-j). \quad (1)$$

The factors in (1) are pairwise disjoint. Call the first m transpositions the A -family and the last m the B -family.

Theorem 1. *For every $m \geq 1$, the permutation $\alpha_m \in \mathfrak{S}_{4m}$ has defect $\delta = 1$ relative to the canonical full cycle γ . Moreover,*

$$\min_{\pi \in [\text{id}, \gamma]} d(\alpha_m, \pi) \geq \left\lceil \frac{m}{2} \right\rceil.$$

In particular, no universal constants θ, κ with the property asked for in the source question exist.

Proof. Since α_m is a product of $2m$ disjoint transpositions, $\#(\alpha_m) = 2m$. Direct evaluation gives the disjoint-cycle decomposition

$$\alpha_m \gamma = \prod_{i=1}^{m-1} (i, 3m-i) (m, 4m, 3m, 2m) \prod_{j=1}^{m-1} (m+j, 4m-j). \quad (2)$$

Thus $\#(\alpha_m \gamma) = 2m - 1$. Since α_m is an involution, $(\gamma^{-1} \alpha_m)^{-1} = \alpha_m \gamma$, and hence $\#(\gamma^{-1} \alpha_m) = 2m - 1$. It follows that

$$\#(\alpha_m) + \#(\gamma^{-1} \alpha_m) = 4m - 1 = (4m) + 1 - 2,$$

which is exactly defect $\delta = 1$.

Now fix $\pi \in [\text{id}, \gamma]$ and write $r = d(\alpha_m, \pi)$ and $\tau = \alpha_m^{-1} \pi$. If the nontrivial cycles of τ have lengths $\ell_1, \dots, \ell_s$, then

$$r = \sum_{q=1}^s (\ell_q - 1), \quad |\text{supp}(\tau)| = \sum_{q=1}^s \ell_q \leq 2r,$$

because $s \leq r$. Every point belongs to exactly one transposition cycle of α_m . Hence at most $2r$ of its $2m$ transposition cycles meet the support of τ . If neither endpoint of such a transposition meets that support, then $\pi = \alpha_m \tau$ agrees with α_m at both endpoints, so that transposition remains a cycle of π . Therefore π contains at least $2m - 2r$ of the displayed transposition cycles.

Within each of the A - and B -families the chords are nested. For every $1 \leq i, j \leq m$, however, their four endpoints occur in the order

$$i < m + j < 3m + 1 - i < 4m + 1 - j,$$

so the i th A -chord crosses the j th B -chord. A geodesic permutation $\pi \in [\text{id}, \gamma]$ is noncrossing and therefore cannot contain a retained A -transposition and a retained B -transposition simultaneously. It contains at most m retained transpositions in total. Consequently

$$2m - 2r \leq m,$$

which gives $r \geq m/2$ and thus the stated integer lower bound.

Finally, if the source conclusion held, then for this defect-one family it would give $d(\alpha_m, \alpha'_m) = 2\delta'_m \leq 2\theta$. Choosing an integer $m > 4\theta$ contradicts the lower bound. The proposed fibre estimate is therefore moot: the required bounded-distance choice does not exist. $\square$

4 Verification

The accompanying script `code/verify_family.py` verifies (2), the defect, and all crossing assertions for $1 \leq m \leq 50$. It also exhaustively enumerates the two smallest cases: for $m = 1$ the nearest geodesic distance is 1, and for $m = 2$ it is 2. These computations only guard against indexing errors; Theorem 1 is entirely symbolic.

5 Scope and novelty check

This disproves the exact full-cycle mapping question on page 148. It is strictly stronger than the separate two-cycle obstruction previously found in this run. A bounded search on 19 July 2026 used the exact source wording and combinations of *distance to noncrossing permutations*, *genus one permutation*, and *Cayley distance*. We also inspected arXiv:1306.4628, arXiv:1502.07674, and arXiv:2110.00176. No source explicitly answering Lancien's question or stating this family was found. Novelty is therefore plausible but provisional pending expert bibliographic review.

6 Human-review recommendation

High priority. The reviewer should check the indexing in (2) and the elementary support argument converting Cayley distance r into at least $2m - 2r$ retained transpositions. No external deep theorem enters beyond the standard identification of $[\text{id}, \gamma]$ with noncrossing permutations.

References

- [1] C. Lancien, *High dimension and symmetries in quantum information theory*, PhD thesis, arXiv:1607.06843 (2016), Section 9.16.3.
- [2] H. Tracy Hall, *Meanders in a Cayley graph*, arXiv:math/0606170 (2006), Theorem 3.1.